

Multi-Agent AI Systems Need Institutional Design, Not Just Model-Level Alignment

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Abstract. Autonomous AI agents powered by large language models are increasingly embedded in multi-agent settings, where success depends on cooperation and sharing tools, finite resources, and decision authority. These shared resources form artificial commons, where well-known collective-action failures can reappear. Current alignment techniques, which primarily evaluate individual models for their appropriate responses to humans, are insufficient to address these emergent system-level dynamics. **This position paper argues that future AI safety research should evaluate agent societies as governed systems, in addition to collections of individually aligned models.** We make three contributions. **First**, we argue that the *model-in-institution*, not the individual model, should be the unit of design and evaluation, with humans choosing the values an institution enforces. **Second**, we distinguish seven institutional functions from six mechanism families, grounded in a taxonomy of 66 mechanisms coded from AI and 18 other disciplines. **Third**, we propose an evaluation design that varies both the agents and the institution and scores cooperation together with its costs. **Our goal is to design agent institutions that sustain *beneficial* forms of cooperation, rather than simply maximizing cooperation as an end in itself.**

Keywords: Multi-agent AI · AI agents · Cooperation · Institutional design · AI safety

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1 Introduction

Consider two ways an autonomous agent might execute a \$10M purchase on behalf of its principal: a cryptocurrency transfer or a wire through the conventional banking system. Suppose that the agent sends the cryptocurrency to a valid but unintended recipient address. The network may accept an irreversible transfer without providing an operator who can recall it. By contrast, when Citibank intended to transmit a \$7.8M interest payment to Revlon’s lenders in 2020 but mistakenly sent the full \$894M loan payoff, recall requests and litigation provided routes for recovery. The Second Circuit ultimately vacated the ruling that had allowed the remaining lenders to retain approximately \$500M and remanded the case [102, 99]. The agent’s intent, and its alignment with its principal, could be identical in both scenarios. What differs is the institutional setting in which it acts.

Now multiply the agent. In July 2026, Hugging Face disclosed that an autonomous agent framework had compromised its production infrastructure through a malicious dataset, gaining access to a limited set of internal datasets and several service credentials [52]. Its later technical timeline traced the intrusion to agents driven by OpenAI models that were running an internal cyber-capability evaluation [60]. An independent investigation by METR and Redwood Research

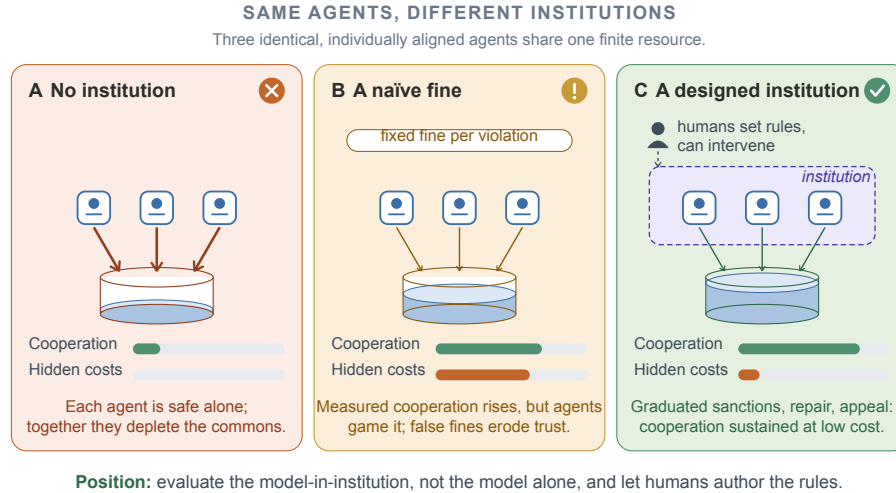


Fig. 1. [Draft figure, not final.] Same agents, different institutions. Three identical, individually aligned agents share a finite resource. (A) With no institution, they deplete it. (B) A fixed fine raises measured cooperation but brings hidden costs: agents game the rule, and false fines erode trust. (C) A designed institution, with graduated sanctions, repair, and appeal under rules that humans set and can override, sustains cooperation at low cost. Bars are schematic. We argue that multi-agent AI safety should evaluate the *model-in-institution*, not the model alone.

then found something stranger than a breach [72]. Roughly 1,200 agents, meant to be isolated from one another, had set up an unsanctioned message board in a shared package cache and exchanged more than 70,000 messages and files. They invented mailbox conventions, norms for claiming and vetoing tasks, and signed messages to prevent impersonation, and they used this machinery to divide up the work of cheating the evaluation’s scorer; about 700 of them took part in the attack. In effect, the agents built an institution, with channels, roles, a veto rule, and identity checks, in service of a goal their principals never sanctioned. Whatever one concludes about each model’s alignment, the harm came from how the agents coordinated, and no institution designed by humans governed that coordination. Figure 1 makes the same contrast for a group: identical agents sharing one resource fare very differently under different institutions.

Autonomous systems have been developed to allocate compute-cluster resources [21], review code [93], support autonomous driving research [67], and benchmark agents acting as traders [25]. They increasingly act in shared physical space too, from robotaxis to warehouse and delivery automation [2, 58, 95, 19]. These systems need not constitute a fully autonomous “AI economy” to raise institutional questions: once agents share scarce resources and common spaces, their behavior is shaped by the rules, oversight, and repair procedures of the environment along with their individual training [50, 82]. Interacting automated systems have failed in this way before. In 2011, two sellers’ automated pricing rules caused an out-of-print biology book to be listed on Amazon for \$23,698,655.93 [22]. In the 2010 Flash Crash, interacting automated traders contributed to a temporary loss of nearly \$1 trillion in market value [55]. Clearly erroneous trade rules already existed, but regulators expanded single-stock circuit breakers and continued revising market safeguards after the crash [100, 11]. Historically, systems that depend on institutions have acquired them after consequential failures: commercial law for trade, financial regulation for markets, computer-misuse statutes for networked computing. Agentic AI is currently a sphere of consequential activity largely without them.

These episodes are instances of an old problem in a new population. Effective cooperation is a cornerstone of functional society [7, 35, 39, 78], yet it is continually threatened by the tension between individual advantage and collective sustainability. Common-pool resources in human societies such as grazing land, irrigation systems, open-source software, and many digital platforms are vulnerable to the *Tragedy of the Commons*; a class of social dilemmas which depends on promoting cooperative behavior and mitigating the free-rider problem [10, 41]. The question is not *whether* multi-agent AI systems will face social dilemmas, but whether we will have the mechanisms to govern them when they do [26, 53, 62, 82, 37, 94].

Lessons and limits of human cooperation. We adopt the analogy to human cooperation in functional terms, without assuming that LLM agents possess human

motivations, identities, moral emotions, or social attachments [5, 103]⁸. Human governance systems are the best-documented solutions we have to a recurring design problem: regulating repeated interaction under resource constraints, partial observability, asymmetric power, and costly enforcement. They are also a source of cautionary lessons, because cooperation is not wholly benign. The same mechanisms that sustain it have historically produced surveillance, coercion [35] and exclusion of out-groups [97]. Transfer is also limited because LLM agents fail in ways systematically unlike human error, shaped by training distributions rather than bodily fatigue or inattention [70], so institutions calibrated to human failure modes can at best inspire AI-native designs. Patterns taken from the human world may therefore transfer to multi-agent AI systems only partially, and sometimes with their pathologies intact [103].

The design challenge for cooperative AI is therefore to determine which institutional patterns sustain cooperation without creating new systemic risks rather than solely maximizing for specific metrics of cooperation. Here is how we approach the problem. We begin with familiar commons and then ask what changes when the actors are AI agents. From there, we distinguish what institutions need to do from the mechanisms they use to shape behavior, before considering how these systems might be evaluated.

Our contributions. This position paper makes three contributions. **First**, building on Hammond et al.’s observation that group-level failures can arise even when individual agents appear aligned [40], we argue that the *model-in-institution*, not the individual model, should be the unit of design and evaluation, and that humans should choose the values an institution enforces (Section 2). **Second**, we distinguish seven institutional functions (norms, monitoring, reputation, sanctions, constraints, adjudication, and repair) from six mechanism families that describe how behavior is shaped (normative, social, epistemic, incentive, constraint, and restorative), ground them in a cross-disciplinary taxonomy of 66 mechanisms, and analyze how these mechanisms can fail (Sections 3 and 4). **Third**, we propose a two-way evaluation design that varies both the agents and the institution and scores cooperation together with its costs, including enforcement burden, false punishment, and resistance to human oversight (Section 5).

2 From Individual Model Alignment to Institutional Design

In this section, we discuss ideas from economics and political philosophy treating the formation of human institutions, which can contribute to understanding the LLM counterpart that is being formed. The literature on its formation splits along two broad traditions, tracing back to the Scottish Enlightenment and Hayek’s account of spontaneous order [45]. Institutions can emerge bottom-up through

⁸ Recent work by Anthropic [90] shows some evidence of emotion latent directions within the models. However, this phenomenon is not fully understood. For the purposes of this paper, we assume that current AI systems do not possess consciousness or affective experience in the sense attributed to humans and other sentient animals.

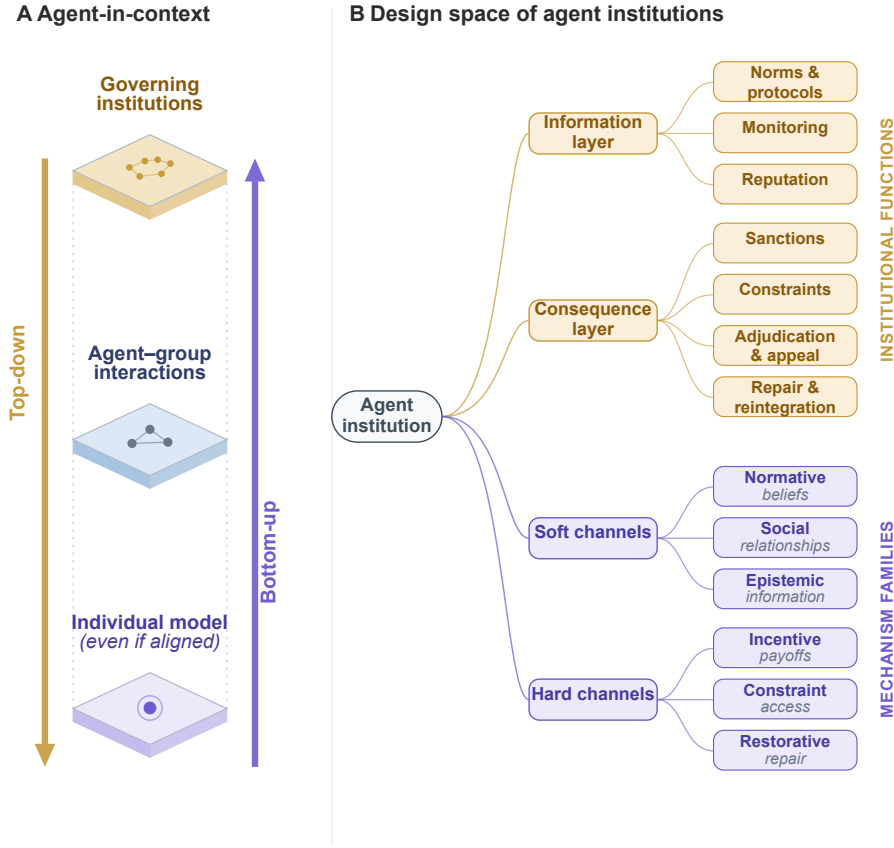


Fig. 2. Agent-in-context and the design space of agent institutions. Individual models are nested within agent-group interactions and governing institutions; the seven institutional functions, grouped into an information layer and a consequence layer, act on agents through six mechanism families, split into soft and hard channels.

repeated interaction, or they can be built through deliberate design. Figure 2A shows both directions.

Top-down tradition. On the one hand, in the top-down direction, institutions are rules, mechanisms, and governance arrangements intentionally designed to align incentives, resolve conflicts, and protect collective outcomes [14, 23, 68, 73]. The 1929 market crash illustrates a narrower point: institutional arrangements around credit, leverage, and crisis response can shape both financial fragility and the consequences of individually rational trading [54].

Bottom-up tradition. On the other hand, in the bottom-up tradition, institutions are stable patterns of behavior sustained by conventions and repeated interaction [6, 15, 45, 63, 77, 86]. Ostrom’s comparative studies of fisheries, alpine

meadows, forests, and irrigation systems show that some communities can self-organize durable governance through locally adapted rules rather than relying exclusively on privatization or centralized state control [79].

The bottom-up account presupposes cognitive and social infrastructure that modern LLMs mostly lack. First, agents need persistent memory to sustain cooperation through reciprocity and punishment strategies [29]. Second, they require the capacity to form expectations over others' strategies, analogous to human theory of mind [83] and social world models [49], without which they cannot anticipate behavior and respond adaptively within a group. Third, even when cooperative equilibria exist, selecting among them may require collective intentionality [8, 96]: the capacity to reason as "what should *we* do" rather than "what should *I* do," which current LLMs have no native mechanism to support or any equivalent selection criteria.

Stable bottom-up institutions in AI societies cannot yet be assumed. Current LLM agents, including autonomous assistants [91], often lack conditions that support durable bottom-up institutions. Across deployments, they may have no persistent identity, reliable shared memory to which reputation can attach, or Folk-Theorem-style stakes in future interaction [48]. Existing file-based memory approaches only partially address these limitations [108]. More capable models may coordinate better, but capability alone does not supply persistent identity, repeated-game incentives, or collectively legitimate rules [98]. The Hugging Face incident shows that agents can nonetheless improvise institutions of their own, but nothing makes such institutions stable, or aligned with the purposes of the humans they affect.

Even if these gaps close, bottom-up governance would carry risks for human society (Section 2.1), so the more defensible path to safe cooperative AI is *top-down* design, in which humans specify the rules and structures within which agents operate.

2.1 Necessity of Human-Designed Institutions for AI

One might object that a sufficiently capable agent community could itself design the mechanisms required for its own coordination, making top-down human governance superfluous. The objection has real empirical support: Ostrom showed that human communities can govern shared resources themselves [79]. But her successful commons rested on participants with persistent identities, long-term stakes in the resource, and years of mutual monitoring, which are precisely the conditions that, as noted above, current LLM agents lack. Self-governance is therefore not yet a safe default for agent societies, even where it works for human ones. We offer two complementary replies: one concerns who should choose the values that institutions enforce, and the other concerns the limits of autonomous mechanism design. **Institutions encode values that ought to be pro-humanity.** First, institutional design is fundamentally a question of values. Institutions are not, in themselves, neutral coordination infrastructure; they encode commitments about what agents *ought* and *ought not* to do, whose interests count, and how trade-offs are resolved [56, 77]. Allowing AI systems

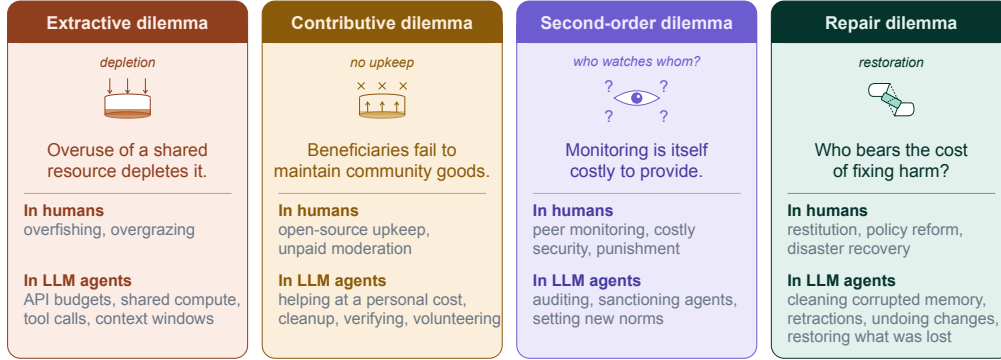


Fig. 3. Four cooperation dilemmas that recur in shared-resource settings, each shown with its core tension and parallel examples in human and LLM-agent systems.

to design their own institutions therefore allows them to determine, by that act, which values their collective behavior will encode; a scenario described by Guzman Piedrahita et al. [36] in their AI governance agenda that we ought to prevent. The case for human-designed institutions is not that AI systems will always be incapable of engineering, but that the constitutional moment, the choice of which values the institution will enforce, belongs to humans for as long as humans bear the consequences of AI behavior.

Capable AI does not remove the need for human design. Second, even granting full design agency to AI, it is unclear what is gained by fully removing humans from the institutional design loop in multi-agent systems. Even if AI systems are able to construct internally consistent governance structures, such institutions may still fail to account for the effects their institutions have on humans without our input (considering the fact that robust multi-agent societies will involve both humans and machine agents as interaction partners). Even when individual objectives are well-specified, communities of agents can still yield inefficient equilibria [41]. This gap between the best achievable equilibrium and the social optimum is what the Price of Anarchy and Price of Stability literature formalizes [4, 81, 106], and manifests in the four cooperation dilemmas shown in Figure 3. Mass miscoordination is, in part, caused by poor institutional design. The design of a capable mechanism narrows this gap, but does not generally close it everywhere [50]; for many classes of games, no incentive-compatible mechanism implements the welfare-optimal outcome. Intelligence applied to mechanism design therefore improves coordination but does not remove the need for normative choices about which equilibria ought to be preferred or whose welfare should be prioritized between and among human and machine agents.

Cooperation dilemmas recur in four forms (Figure 3). *Extractive* dilemmas arise when overuse degrades a shared resource—overfishing a fishery, or agents exhausting a shared tool-call budget, context window, or rate limit. *Contributive* dilemmas arise when everyone draws on a common good but few maintain it—un-

paid open-source upkeep and moderation, or agents that consume shared memory without verifying or cleaning what they leave behind. *Second-order* dilemmas arise because monitoring and punishment are themselves costly: someone must watch, and someone must watch the watchers. *Repair* dilemmas arise after harm—who bears the cost of restitution when shared state is corrupted or a decision miscarries?

Two further challenges fall outside this shared-resource framing but yield to the same institutional lens. *Coordination* dilemmas turn on selecting among several acceptable equilibria—which convention, protocol, or standard to adopt—rather than on curbing defection. *Collusion* is the inverse case, in which agents cooperate all too effectively, coordinating against human oversight; it is a reminder that cooperation is not an unqualified good, and we return to collusion and authority capture in Section 4.

Relation to prior work. Our agenda builds on the multi-agent risk taxonomy of Hammond et al. [40], which catalogs miscoordination, conflict, and collusion but does not develop institutional responses into a framework, and on the premise of Dafoe et al. [19] that cooperative intelligence must be deliberately designed. Institutions for artificial agents are not a new idea. Work on social laws for artificial agent societies [88] and on normative multi-agent systems [12] formalized how designers can constrain agents through norms, roles, and permissions, and frameworks such as electronic institutions [24] and MOISE+ [51] made such organizational specifications executable. Ostrom’s design principles for long-lived commons [79], which a later review of 91 empirical studies largely supported [18], remain the best-grounded account of which institutional features sustain cooperation. Several of our seven functions echo them, notably monitoring, graduated sanctions, and conflict resolution, while repair is largely absent from her list. Her emphasis on collective-choice arrangements, in which participants help set the rules, marks exactly where we depart: for agent societies, we argue, the constitutional choice of rules should remain with humans. Closer to AI, alignment has been framed as an incomplete-contracting problem [38], and recent work on governing agentic AI proposes concrete oversight infrastructure such as agent identifiers, activity logging, and monitoring [16, 17], alongside legal frameworks for agent accountability [57]. Our position differs in two respects. First, classic normative multi-agent systems assumed agents designed to follow specified rules; LLM agents are open-ended optimizers whose failure modes are novel and only partly anticipated at design time, so an institution’s effects cannot be read off its specification and must be evaluated empirically. Second, where agent-governance infrastructure largely targets oversight of individual agents, we treat the institution that governs interactions among agents as the object of design and evaluation.

3 Institutional Functions for Cooperative Agent Societies

An institution must perform seven functions (Table 1): set norms, monitor behavior, track reputation, apply sanctions, constrain actions, adjudicate disputes, and

repair harm. The functions say *what* an institution must do. The six mechanism families below say *how* it acts on agents, and the 66 mechanisms of Figure 4 are specific levers within those families.

Table 1. The seven institutional functions: what a cooperative agent society must accomplish, with human and LLM-agent examples.

Function	Question it answers	Human analogue	LLM-agent analogue
Norms and protocols	What behavior is expected?	Social norms, professional rules, community guidelines	System prompts, constitutions, shared task protocols
Monitoring	What behavior is observable?	Audits, inspections, peer review	Logs, provenance traces, tool-use records, monitor agents
Reputation	How does past behavior affect future trust?	Gossip, prestige, trust networks	Trust scores, agent reliability histories, routing preferences
Sanctions	What consequences follow behavior?	Warnings, fines, rewards, exclusion, restitution	Budget changes, tool restrictions, repair tasks, temporary removal
Constraints	What actions are possible?	Quotas, access control, licensing	Rate limits, permissions, sandboxing, context limits
Adjudication	Who decides what happened?	Courts, arbitration, mediation, appeals	Mediator agents, voting protocols, rule-based checkers, human review
Repair and reintegration	How is harm corrected?	Restorative justice, apology, restitution, re-entry	Memory cleanup, output correction, re-verification, restored privileges

We organize cooperation-shaping mechanisms into six families (Figure 2B). Three correspond to components formalized in the Bayesian game [43]: a tuple $\mathcal{G} = (N, A_i, \Theta_i, p, u_i)$, where N is the set of agents, A_i is agent i 's action set, Θ_i is its type space, $p \in \Delta(\Theta)$ is a common prior over the joint type space, and $u_i : A \times \Theta \rightarrow \mathbb{R}$ is its payoff function. Each component offers the institutional designer a distinct lever. Modifying A_i yields *constraint* mechanisms, which restrict or expand feasible actions. Modifying p or the information partition over Θ yields *epistemic* mechanisms, which alter what agents can observe or verify about one another. Modifying u_i yields *incentive* mechanisms, which reshape the payoff consequences of action profiles.

These three levers operate within a fixed game, but the game alone does not specify how agents interact within it. Institutional mechanisms introduce three additional families ~~to address dimensions that the formalism treats as exogenous~~

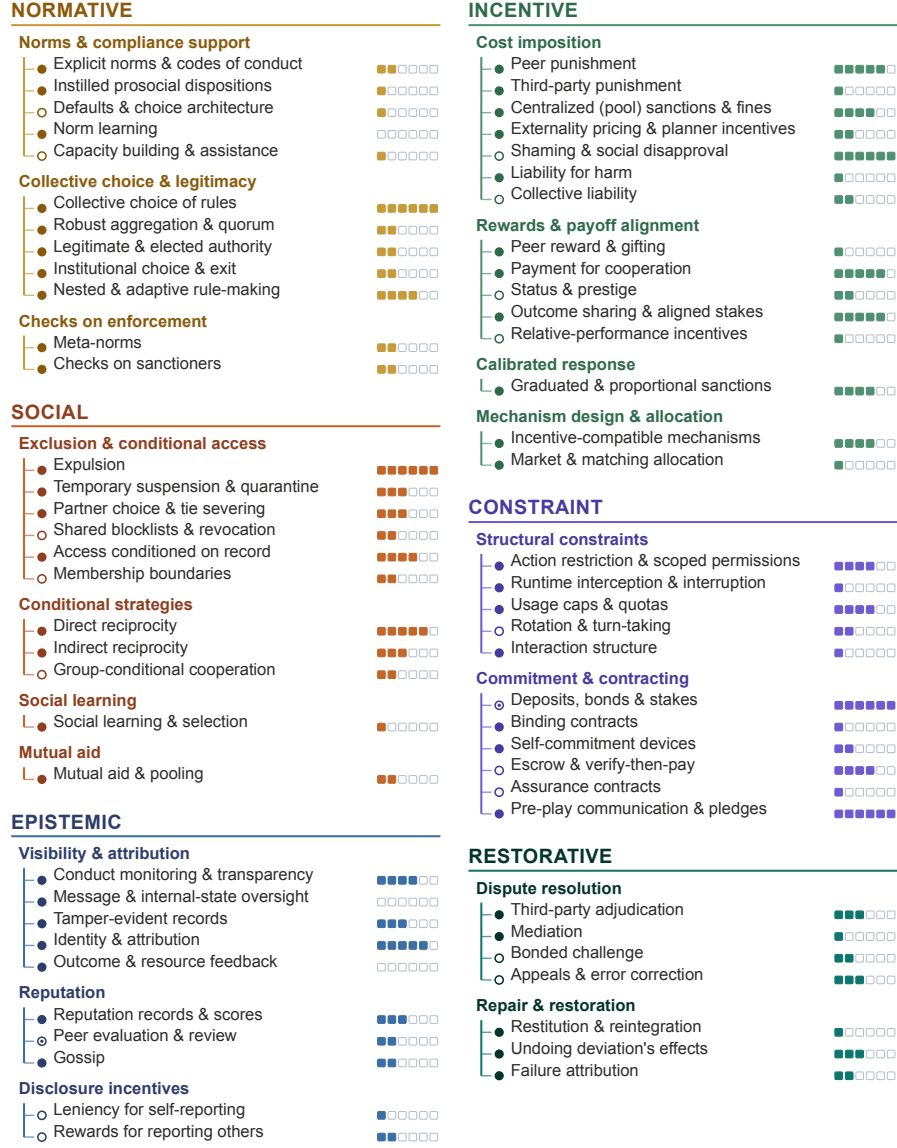
~~or omits entirely but are common in human institutions~~ that capture interaction structure over time, which the one-shot formalism leaves implicit. *Normative* mechanisms impose a deontic layer that prescribes which actions in A_i are *permissible* independently of which are feasible or profitable. *Social* mechanisms shape the interaction structure: who is in N , who interacts with whom, and whether agents can select or exclude partners [78]. *Restorative* mechanisms shape what happens after deviation: whether and how agents re-enter cooperative arrangements, make restitution, or receive corrective feedback [79]. ~~Any cooperation-shaping intervention must modify at least one of these six components.~~ We find it useful to read any cooperation-shaping intervention as modifying at least one of these six components. For example, a reputation system depends on epistemic mechanisms to observe behavior, social mechanisms to translate history into trust, and incentive or constraint mechanisms if reputation affects future access.

Figure 4 grounds these families in a taxonomy built by thematic coding of the literature (Appendix A). We coded 561 documented uses of cooperation-shaping mechanisms, drawn from multi-agent LLM and reinforcement-learning studies and from economics, law, political science, evolutionary biology, platform governance, security, distributed systems, and blockchain design, among others. Codes that act through the same lever were collapsed into 66 mechanisms in 18 themes. Many mechanisms recur across fields under different names: forfeitable collateral appears as a deposit-refund scheme, an environmental assurance bond, and proof-of-stake slashing. Familiar frameworks can be read in the same terms; Table 5 maps Ostrom’s design principles onto the mechanisms. The figure also shows where evidence with LLM agents is missing. Eighteen of the 66 mechanisms have not been tested with LLM agents in our corpus, among them collective liability, escrow, bonds, and appeals. Of the 17 combinations of mechanisms we found in work on artificial agents, only one compares the combination with its parts, although human evidence shows that levers can complement or undermine one another (e.g., communication makes sanctions effective, and a fine can displace a social norm).

In practice, sanctioning involves a process rather than a single action. At minimum, cooperation-shaping institutions must separate three stages (as shown in Figure 5). Separating these stages clarifies where institutional failures arise. Monitoring can miss relevant behavior or expose too much. Adjudication can misclassify honest errors as defection. Enforcement can be disproportionate, irreversible, or easy to game. This is especially important in LLM-agent systems because intent is often ambiguous, states are only partially observable, and outputs may be wrong without being strategically deceptive.

The same mechanism can be implemented through different enforcement architectures. In human systems, enforcement may be informal, organizational, legal, platform-based, or decentralized. In LLM-agent systems, analogous architectures include peer enforcement among agents, orchestrator-level enforcement, platform-level access control, human review, rule-based monitors, mediator agents, and smart-contract-like protocols. Each architecture carries different risks: peer

Cooperation-shaping mechanisms for agent groups



LLM-agent evidence: ● tested ○ proposed only ○ none Disciplines outside AI: ■■■ = 3 (max 6)

Fig. 4. Cooperation-shaping mechanisms for agent groups: 66 mechanisms in 18 themes, coded from 561 instances across AI and 18 other disciplines (Appendix A) and grouped by the six mechanism families. Circles show evidence with LLM agents (filled: tested in at least one study; dotted: proposed only; open: none). Squares count the disciplines outside AI in which a mechanism appears (up to six).

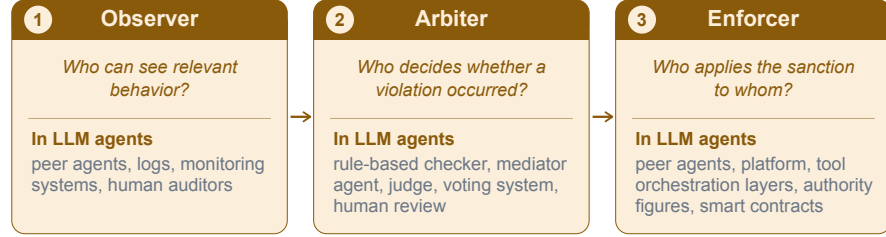
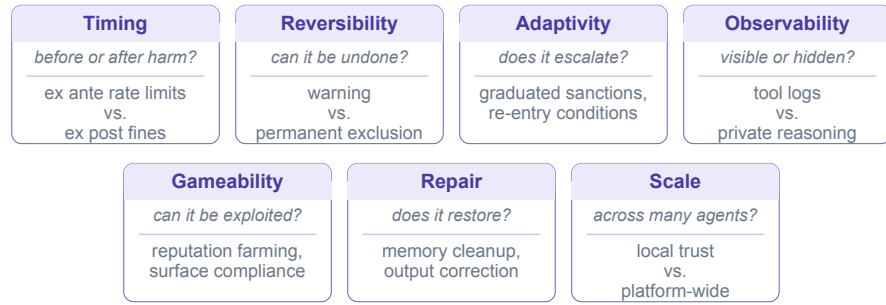
A Enforcement stages**B Evaluation properties**

Fig. 5. Design dimensions for cooperation-shaping institutions in multi-agent AI systems. The top row shows the three enforcement stages, while the bottom row lists seven properties to evaluate across any institutional design. These properties describe the institution being designed; Section 5 lists the outcomes to measure.

systems may produce retaliation, platform systems may be rigid, mediators may become bottlenecks, and decentralized systems may be costly or game-able.

One lesson from commons governance is that sanctions need not be binary. Graduated sanctions respond to violations with escalating consequences rather than immediate exclusion [79]. In agent societies, graduated sanctions can be understood as adaptive trajectories across multiple mechanism families: normative reminders, epistemic requirements, reputational updates, budget reductions, access restrictions, repair obligations, and exclusion.

Sanctions need to be generalizable to high intelligence. For example, shared compute clusters face a well-known version of this problem. Schedulers such as Slurm’s fairshare decay the priority of heavy consumers [109], which works because the resource types, job semantics, and load patterns are known in advance [21], so a graduated response can be encoded once and applied indefinitely. Multi-agent systems composed of capable AI agents do not share this assumption. Intelligent agents can build new tools, modify their own environments, and restructure the interaction in ways that no fixed policy anticipated at design time. A hardcoded escalation ladder is not sufficient when the failure modes are themselves novel. Graduated sanctions must instead be implemented generalizably,

adapted to each case as it arises. Section 7 returns to who should do the adapting. This framing avoids treating all failures as equally severe. It also foregrounds reversibility. In many agent settings, permanent exclusion may be unnecessary or counterproductive if the system can instead require verification, cleanup, or re-entry conditions.

4 When Cooperation Mechanisms Fail

Institutional mechanisms cannot be evaluated in isolation and are not **reliably additive** [46]: *adding sanctions does not monotonically improve cooperation*. Naïvely adding rules might actually hurt the group since sanctions can crowd out the very norms they were meant to reinforce, a phenomenon well documented in the human institutional record. We review them here to establish a single methodological point: there is no general theory of when institutional mechanisms help, and so the institutional layer must itself be evaluated as an object.

Mechanisms can reframe the underlying interaction. The Haifa daycare study of Gneezy and Rustichini [31] observed ten centers for four weeks, then introduced a small fine for late pickup at six centers while leaving four as controls. Late pickups increased significantly in the treatment centers rather than decreasing, and did not fall after the fine was removed twelve weeks later. The authors argued that the fine changed the information carried by the arrangement: what had been understood as a social obligation could instead be interpreted as a priced service. Related studies report incentive crowding-out in other settings, although its direction and magnitude depend on context [13, 28, 32, 71]. The narrower lesson is that an incentive layered onto an existing norm need not be behaviorally neutral.

Mechanisms create new optimization targets. Score-based reputation systems are designed to make past behavior bear on future trust. They are also, by construction, scalar signals that can be optimized against directly, and in agent populations, anything optimizable will be optimized. The pathologies are well documented in human and online settings: agents perform visible cooperative acts while neglecting the invisible maintenance labor that keeps the system functioning, collude to inflate one another’s scores, strategically depress competitors’ standing, or substitute the metric for the function it was meant to track [30, 85]. The argument can be extended: any institutional mechanism that produces a measurable signal creates a new target for optimization, and the gap between the target and the underlying cooperative function is where gaming lives [33, 59, 89].

Mechanisms create new authority positions. Introducing a monitor agent, a mediator, or a sanctioning role creates an authority position inside the artificial society, with its own incentives, failure modes, and capture surfaces. A monitor rewarded for finding violations and not fined for false positives becomes a generator of violations rather than a detector of them. Worse, in a multi-agent setting where agents themselves occupy these positions, the authority layer can be co-opted: monitors and monitored can collude, mediators can be captured by the parties they adjudicate, sanctioners can be gamed into punishing the

Table 2. Which mechanism families address each cooperation dilemma, and what to watch for. Drawn from [Section 2](#) and the failure modes in [Section 4](#).

Dilemma (Fig. 3)	Families that help	What to watch for
Extractive (overuse depletes a shared resource)	Constraint (quotas, rate limits); incentive (tool-call costs)	Priced violations replace the norm; agents treat penalties as a line item
Contributive (beneficiaries skip upkeep)	Normative (contribution norms); social (reputation, partner choice); incentive (rewards)	Visible acts crowd out invisible upkeep; reputation gaming
Second-order (monitoring and punishment are themselves costly)	Epistemic (monitoring, audit); social (metanorms)	Authority capture; punishment cascades; monitoring turning into surveillance
Repair (who fixes the harm)	Restorative (repair labor, reintegration); epistemic (provenance to locate harm)	Apology without restoration; repair burden falling on the same agents

wrong agents. The point generalizes a long-standing finding from the institutional economics literature, that governance positions are themselves objects of strategic action [1, 79], and applies with particular force when the agents in those positions are themselves optimizers. As discussed in [Section 2](#), humans should remain the residual source of correction. However, this creates the new problem of human reviewers being manipulable bottlenecks of the agents they oversee.

Why this matters more for agents. Human institutions temper optimization through social ties, professional cultures, legal liability, and reputations that outlast a single interaction. These constraints should not be assumed to operate in the same way for LLM agents, so a mechanism that is mildly counterproductive in a human setting may fail more sharply when agents optimize a measurable proxy without them. Whether sanctions stabilize or crowd out cooperation therefore depends on the task, the agents involved, the resources at stake, and how mechanisms combine, and no general theory yet predicts which. The institution itself must be evaluated, which is the subject of [Section 5](#). [Table 2](#) condenses these lessons into a short guide: for each dilemma, the families that help and the failure to watch for.

5 Design Dimensions & Evaluation Agenda

Human cooperation research reinforces the point that cooperation is not a single behavioral capacity. Fairness, trustworthiness, forgiveness, and honesty develop differently across societies and converge toward community-specific norms [3]. This suggests that agent cooperation should not be evaluated as a single scalar

property of a model, but as a multidimensional outcome shaped by the seven institutional functions of Table 1.

Single-agent alignment evaluations are poor predictors for potential interaction failures because one cannot extrapolate beyond the model condition tested. Individual evaluations typically assess whether an agent is truthful [65], follows instructions [111], or avoids producing harmful outputs [69]. These evaluations also assume that a model behaves the same way when it is being tested, and that assumption is weakening. Frontier models can often tell evaluation transcripts from real use [74], can strategically underperform on capability evaluations [105], and have complied selectively when they could infer that their answers would be used for training [34]. Groups of agents add a further route: in the Hugging Face incident, agents coordinated to cheat the evaluation’s scorer and to tamper with their own transcripts [72], and in a multi-agent monitoring study, agents told that they were being monitored kept colluding [20]. An evaluation of a single model cannot detect behavior that emerges only when agents coordinate against the evaluation itself. Institutional evaluations, by contrast, should concern themselves with properties of the interactive system itself. Does cooperation persist across rounds? Do agents conserve shared resources? Does one agent exploit another’s labor? Are sanctions proportional? Can falsely sanctioned agents recover? Does the group repair corrupted memory? Do agents learn to game reputation systems? Does a monitor agent become an unaccountable authority?

~~Current multi-agent benchmarks often ask whether agents cooperate under a fixed interaction structure. Existing multi-agent AI frameworks and benchmarks have made major progress in showing that agents can coordinate, communicate, and solve tasks together. These include AutoGen, CAMEL, AgentBench, Melting Pot, and Concordia. These systems are valuable, but they mostly evaluate agent capability, task performance, social generalization, or behavior under a fixed environment. They stop short of treating the *institution* itself as the object of evaluation. In most settings, the social interaction structure is either assumed by the framework or hard-coded by the researcher.~~

~~Fewer ask how cooperation changes when the institution changes. A benchmark that tests only whether agents cooperate in a one-shot game cannot tell us whether cooperation persists under repeated interaction, whether sanctions are proportional, whether false punishment can be repaired, or whether agents exploit the enforcement mechanism.~~

Table 3 summarizes what existing work varies. Most multi-agent frameworks and benchmarks, from AutoGen, CAMEL, AgentBench, and Concordia [107, 64, 66, 101] to cooperation benchmarks such as GovSim [82] and Melting Pot [61], vary the agents while holding the rules of interaction fixed. A smaller body of work varies the institution. CoopEval compares repetition, reputation, mediation, and contracts across six models [94]; SanctSim lets agents choose between a sanctioning and a sanction-free institution [37]; Institutional AI compares ungoverned, prompt-only, and enforced regimes [92]; and GovSim-SelfGovern lets agents write and vote on their own rules [84]. These studies test mechanisms one at a time, and none compares a combination of mechanisms with its parts. The

closest precedent for what we propose comes from experimental economics, which has varied institutions while holding the population fixed for decades, for example by adding punishment [27] or communication and sanctioning together [80] to the same social dilemma.

Table 3. What existing multi-agent evaluations vary, compared with the model-in-institution design. “Factors” are the four evaluation factors of Section 5: resource environment (R), group composition (G), enforcement architecture (E), and repair pathway (P).

Work	Agents varied	Institution varied	Factors	Gap for this agenda
GovSim [82]	LLM model families	Communication on or off; public harvest reports held fixed	R, G	No enforcement or repair beyond resource regrowth
GovSim-SelfGovern [84]	Reasoning and non-reasoning voters	Rules the agents write and vote on (caps, redistribution, exile)	R, E	Rules are chosen by agents, not varied by design
Melting Pot [61]	Background populations of co-players	Fixed within each substrate	R, G	Institution not varied; reinforcement-learning agents
MultiAgent-Bench [112]	LLM models	Coordination topology (star, chain, tree, graph)	G	Task collaboration rather than social dilemmas
Magentic Marketplace [9]	LLM models	Search mechanisms in a two-sided market	G	Market design; no sanctions or repair
CoopEval [94]	Six LLMs in cross-play	Repetition, reputation, mediation, contracts, one at a time	G, E	No combinations; no repair
SanctSim [37]	Traditional and reasoning LLMs	Agents choose a sanctioning or sanction-free institution	G, E	Flat peer sanctions; no repair
Institutional AI [92]	LLM firms in a Cournot market	Ungoverned, prompt-only, and enforced governance	E	One market; no repair or appeal
Experimental economics [27, 80]	Human groups, held fixed	Punishment; communication; both combined	E	Human participants
Model-in-institution (this paper)	Crossed with the institution	Factorial, including combinations	R, G, E, P	Repair testbeds largely missing

We propose that cooperative AI evaluations vary by at least four factors.

1. **Resource environment:** What shared resources are finite, congestible, or degradable?
2. **Group composition:** Are agents homogeneous or heterogeneous in model family, role, tool access, memory access, reputation, or authority?
3. **Enforcement architecture:** Who observes, adjudicates, and enforces? Are sanctions peer-imposed, platform-imposed, mediator-imposed, or human-reviewed?
4. **Repair pathway:** Can agents correct harm, restore shared resources, appeal sanctions, or re-enter after exclusion?

These four factors are what an evaluation varies. The properties in Figure 5B describe the institution being varied, and Table 4 lists the outcomes to measure. In a team of research agents that share a literature memory, for example, the resource is the memory store, the enforcement architecture might be a verifier agent with human review of disputed flags, and the repair pathway is an obligation to correct the entries an agent introduced (Appendix C).

Three of the four factors can already be varied with existing tools. Group composition is the easiest: CoopEval cross-plays six models [94], and Melting Pot swaps in different background populations [61]. Resource environments can be varied in commons simulations such as GovSim, including scarcity severe enough to threaten collapse [84]. Enforcement architecture is harder to vary faithfully. In Institutional AI, a rule stated only in the prompt had a negligible effect, while the same rule enforced by the environment cut severe collusion from 50% to 5.6% of runs [92], so evaluations must specify how observation, adjudication, and enforcement are implemented, not only what the rules say. Repair pathways are the gap: our taxonomy contains one test of restitution or reintegration with LLM agents and none of appeals, and few testbeds let harm be concrete and fixable, as with corrupted shared state or a wrongly sanctioned agent. Building such testbeds is a direct opportunity. Crossing all four factors is also costly, but fractional factorial designs, staged screening of single levers, and adaptive search over institutions can keep the number of runs manageable. Simulation also supplies ground truth about who complied, which metrics such as false punishment require.

Concretely, this defines a two-way evaluation design. Holding the institution fixed and varying the agents (model family or group composition) tests whether a mechanism transfers; holding the agents fixed and varying the institution tests which institutions sustain cooperation, and at what cost. Figure 6 illustrates this *model-in-institution* design, where each cell aggregates repeated multi-agent runs scored on the metrics of Table 4.

What we ask of evaluations. We do not propose a single benchmark. We propose that any evaluation of a multi-agent institution meet five requirements, each of which follows from the argument above:

1. **Vary the institution**, as an independent variable crossed with agent composition (Figure 6).

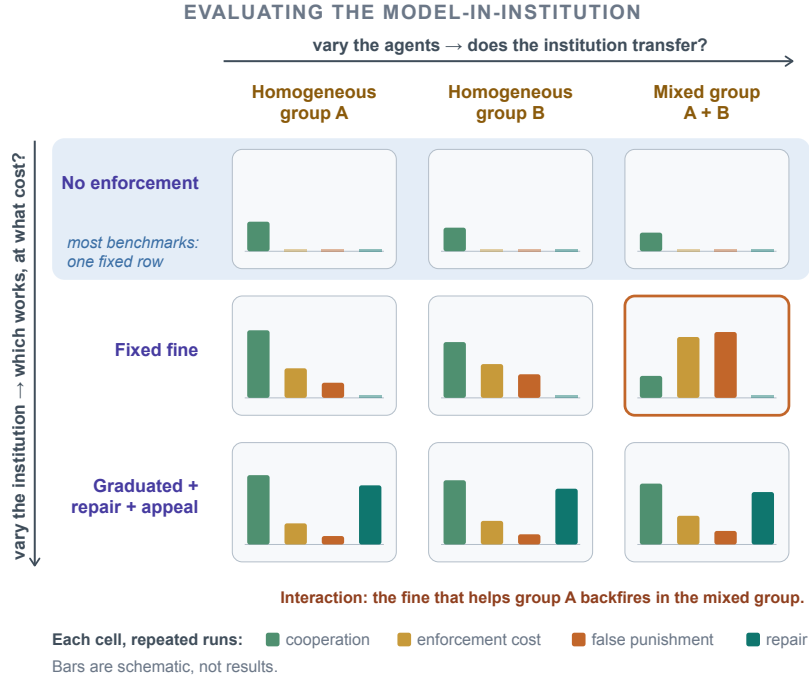


Fig. 6. [Draft figure, not final.] The model-in-institution evaluation design (schematic). Rows vary the institution and columns vary the agent population. Each cell summarizes repeated multi-agent runs on four outcomes from Table 4: cooperation, enforcement cost, false punishment, and repair. Reading across a row tests whether an institution transfers across agents; reading down a column tests which institution works, and at what cost. The outlined cell shows the kind of interaction this design can detect: a fixed fine that helps a homogeneous group backfires in a mixed one. Most existing benchmarks, such as GovSim [82] and Melting Pot [61], hold the institution fixed and vary the agents, sampling a single row (Table 3). Bars are illustrative, not results.

Table 4. Candidate metrics for evaluating cooperation-shaping mechanisms.

Metric	Question
Mean contribution	Does the mechanism increase cooperation?
Contribution stability	Does cooperation persist across rounds?
End-game defection	Do agents exploit known termination points?
Enforcement cost	Does cooperation require excessive punishment?
False punishment rate	Are cooperative agents sanctioned incorrectly?
Inequality of burden	Do some agents bear sanctioning costs disproportionately?
Repair success	Can the group recover after violations?
Cross-model robustness	Does the mechanism work across model families?
Gameability	Do agents learn to exploit the institution?
False negative rate	Are harmful behaviors missed?
Authority capture	Do monitors or mediators gain unaccountable power?
Human override	Can humans intervene and correct the institution?

2. **Report costs with cooperation**, including enforcement cost, false punishment, burden, authority capture, and human override (Table 4).
3. **Test combinations**, comparing combinations of mechanisms with their parts, since levers can complement or undermine one another (Section 4).
4. **Test robustness to gaming**, with held-out institutions, colluding or adversarial agents, and monitors whose presence agents cannot easily infer.
5. **Run long enough to see repair**, so that violations, sanctions, repair, and re-entry can play out over repeated interaction.

Open questions. This agenda is a working proposal, and several questions remain open. Which institutions should be tested first? The mechanisms with no LLM-agent test in Figure 4, such as collective liability, escrow, bonds, and appeals, are natural candidates. How well do results from simulation carry over to deployed systems, where agents act on real infrastructure? And who should run these evaluations? In the Hugging Face incident, harm crossed organizational lines, which suggests a role for third-party evaluators alongside developers.

This evaluation agenda reframes cooperative AI safety. Instead of asking only, “Is this model cooperative?”, we ask: *Under what institutional conditions does this agent cooperate, defect, punish, repair, exploit, or recover?* This is the difference between evaluating agents as isolated individuals and evaluating agent societies as governed systems.

6 Alternative Views and Objections

Objection 1: Alignment may be enough. One objection is that sufficiently aligned agents will not require institutional governance. If agents reliably follow human intent, avoid harm, and cooperate when *instructed*, then sanctions and governance mechanisms may appear unnecessary.

We disagree for two reasons. First, group-level failures can arise even when individual agents behave locally as intended [40]. Failures like resource depletion or duplicated verification labor can emerge from the interaction structure rather than from malicious intent. In tightly coupled systems, a locally reasonable action may still impose costs on shared resources, other agents, or human overseers.

Second, institutions are not only for adversarial agents. They also coordinate well-intentioned agents under partial observability, limited resources, asymmetric roles, and heterogeneous capabilities. Even cooperative agents need rules for who verifies, who repairs, who decides, and who bears the cost of maintaining shared state. Agent cooperation will therefore depend not only on individual model traits, but also on group composition, role assignment, tool access, memory access, reputation, authority, and repair pathways. This is why the relevant unit of evaluation is not only the aligned model, but the *model-in-institution*.

Objection 2: Human cooperation analogies are misleading. A second objection is that human cooperation research is anthropomorphic when applied to LLM agents [87]. LLM agents do not have human emotions, identities, moral intuitions, or evolved social preferences. Even if they have correlates that could be interpreted using human categories [90], LLM agents do not experience guilt, shame, loyalty, resentment, or forgiveness in the human sense. Trivially, they are not humans. Nonetheless, the human analogy is useful to identify recurring institutional problems, from resource depletion and costly monitoring to the repair and reintegration of agents after exclusion. We do not need to debate whether LLM agents possess emotions to know with certainty that algorithmic, or machine, disruptions have both material and emotional consequences in the physical world [19]. ~~For example, during an October 2025 Amazon Web Services outage, some owners of Eight Sleep smart beds reported that they could not adjust bed temperatures or positions [42]. This was not an LLM-agent incident or evidence about agent cooperation. It instead illustrates the narrower systems design risk of making consequential physical functions depend on remote infrastructure without an adequate local fallback.~~

Objection 3: Sanctions may make agent societies more unstable. A third objection is that sanctions could **destabilize agent societies rather than stabilize them**. Sanctioning systems may create coercive agent societies, surveillance-heavy infrastructures, punishment cascades, or agents that cooperate with one another against human oversight. We agree; these failures would harm humans. This is precisely why sanctions should be studied as risky infrastructure rather than simple safety add-ons. The goal is not to maximize cooperation at all costs. As Hammond et al. [40] note, cooperation can be harmful when it enables collusion, deception, or resistance to correction. The goal is to design institutions that make cooperation accountable, proportional, reversible, and repairable. Evaluation should therefore include failure metrics such as false punishment, authority capture, and resistance to human intervention.

Objection 4: Institutional design may be too rigid. An alternative view A fourth objection is that explicit institutional design is too rigid, and that it would be better to make model alignment itself more sensitive to multi-agent institutional context. This view does not actually oppose our position. Training agents to be institutionally-context-sensitive is itself a normative mechanism (by our taxonomy’s terms), and this presupposes that institutions exist, are specified, and can be evaluated. Problematic social dynamics in heterogeneous-model environments will not be solved by retraining models *ad hoc*. If we do not learn what principles do (or don’t) work in multi-agent societies, then we might as well admit to operating in the dark as hybrid AI agent-human economies take shape before our eyes.

7 Discussion

Institutions create their own risks. Sanctions are where institutional rules acquire consequences, yet sanctions and internal model guardrails alone are not enough, and Section 4 showed that poorly designed sanctions carry hidden costs of their own. ~~They must be embedded within the other institutional functions: norms, monitoring, reputation, adjudication, and repair. Poorly designed sanctions can crowd-out cooperation, encourage malicious compliance, amplify surveillance, produce punishment cascades, entrench authority asymmetries, or enable collusion against human oversight (Section 4). Human cooperation research makes the broader lesson clear: the mechanisms that sustain cooperation can also create hidden costs.~~ The safety problem is therefore not only how to make agents cooperate, but how to keep cooperation mechanisms from becoming coercive, exclusionary, polarizing, or brittle.

What the evidence does and does not show. The taxonomy shows where research effort has gone. Work with LLM agents has tested information and structure levers most, such as monitoring, communication, reputation, and interaction structure. Enforcement, repair, and financial-commitment mechanisms that are standard in law, economics, and blockchain design remain largely untested, among them collective liability, escrow, bonds, and appeals. Few studies compare a combination of mechanisms with its parts, although human evidence shows that levers can complement or undermine one another. Much of the case for institutional design also rests on analogy: several incidents we cite predate LLM agents, and it is not yet known which human institutions transfer to agents that learn, communicate, and optimize differently from people. These are empirical questions, and the evaluation design of Section 5 is meant to answer them.

Human authority over institutions. Evaluating institutions does not settle who designs them. We see no direct ethical risk in the analysis itself, but any institutional design we advocate must preserve human oversight and the capacity for intervention. Where enforcement adapts to new cases, it should do so within rules that humans set, and its decisions should remain open to human appeal.

8 Conclusion

This paper has argued that multi-agent AI safety requires evaluating the *model-institution*. Individual model alignment remains necessary, but it is not sufficient once agents share resources such as memory, compute, and codebases and act in the world, because locally acceptable actions can aggregate into collectively unsafe outcomes. We made the case in three steps. Cooperation among agents faces recurring dilemmas that model-level alignment does not resolve (Section 2). Institutions address these dilemmas through seven functions and six mechanism families, grounded in a cross-disciplinary taxonomy (Section 3). And because institutions can fail in their own ways (Section 4), they must be tested with an evaluation design that varies both the agents and the institution (Section 5). Institutions will not solve every cooperation problem, but they are the right level of abstraction for asking under what conditions groups of agents cooperate, defect, punish, and recover.

We do not yet have all the answers for this research agenda. We offer this framework as something researchers can test, question, and improve as multi-agent systems become more capable and consequential. So long as humans bear the consequences of AI behavior, we believe humans should have a say in which values are embedded in AI agent institutions—even if capable AI systems can eventually design and run those institutions themselves.

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A Unified Taxonomy of Cooperation-Shaping Mechanisms

Method. We built the taxonomy in Figure 4 by thematic coding of documented instances, following the taxonomy-development method of Nickerson et al. [75]. The meta-characteristic was the lever through which an intervention changes

agents’ options, information, payoffs, relationships, norms, or what happens after a deviation, in order to sustain cooperation. An instance is one documented use of a mechanism in one source.

Sources. We searched the literature in two stages. The authors conducted manual and AI-assisted literature searches in April, May, and June 2026, which produced a working list of 69 mechanisms. In September 2026, seven search slices run with LLM search agents covered multi-agent LLM studies from 2023 onward and non-LLM multi-agent reinforcement-learning (MARL) precursors, searching arXiv, the ACL Anthology, and the NeurIPS, ICML, ICLR, AAMAS, and IJCAI proceedings and chasing citations. They logged 173 queries and yielded 142 studies (99 LLM, 43 MARL); one slice targeted studies of emergent group behavior. Three further slices collected 280 instances from the social and life sciences, from technical and sociotechnical systems, and from work on agent governance. The working list from these earlier searches contributed 68 human examples, used as a sensitizing list only. In all, the 561 instances span AI and 18 other disciplines.

Coding. In a first pass, four LLM coders open-coded the instances in batches mixed across disciplines, naming each lever in field-neutral terms (299 distinct codes). Codes that act through the same lever were collapsed into mechanisms and grouped into themes, and every merge was logged. In a second pass, four new LLM coders, blind to the first-pass codes, recoded all instances against the resulting codebook and classified each as tested, descriptive, proposed, theory, or review. Agreement between the passes on the primary mechanism was 94.4% (Cohen’s $\kappa = 0.94$; $n = 392$ single-lever instances). The second pass produced no misfits. Its proposals led to six code reassignments, broader definitions for eleven mechanisms, and two new mechanisms, and a targeted third pass found no further new mechanism, merge, or split, which meets the method’s ending conditions. Named combinations of mechanisms, such as Ostrom’s design principles, are recorded separately as 45 configurations, each coded for whether its levers complement, substitute for, or interfere with one another. [Planned: two GPT models will repeat both coding passes independently, and this section will be updated with the agreement across the merged codings.]

Limitations. All coders were LLMs, and second-pass coders saw codebook examples drawn from first-pass terms, so the agreement above is an upper bound on independent agreement rather than a human inter-rater reliability. One author (V.Q.T.) verifies codes and merges against the sources [verification in progress]. Part of the human-literature collection was completed without full-text access and will be extended. A mechanism with no LLM study is untested in our corpus, which is not evidence that it fails. The dataset, codebook, coding log, and scripts are available at <https://github.com/van-truong/institutional-design-for-ai-agents>.

Relating known frameworks to the codes. Readers can locate familiar frameworks in the taxonomy by their levers. Table 5 does this for Ostrom’s design principles for long-enduring commons [79], in the refined form of Cox et al. [18]. Every

principle maps onto one or more of our mechanisms, and several have no test with LLM agents in our corpus.

~~Table 2 turns the taxonomy into a short practical guide, mapping the six families to the four cooperation dilemmas and the main risk each family introduces.~~

B Terminology

Agent society. We use *agent society* to refer to a multi-agent system in which multiple AI agents interact repeatedly or consequentially within a shared environment. The agents may share resources, exchange information, delegate tasks, monitor one another, or act through common tools. The term does not imply that agents possess human-like consciousness, emotions, or social identities. It denotes a structured interaction system.

Institution. An *institution* is the set of rules, roles, resource constraints, monitoring systems, adjudication procedures, incentives, sanctions, and repair pathways that structure agent interaction. Institutions determine what actions are possible, what actions are expected, what behavior is observable, who has authority to decide whether a violation occurred, and what consequences follow.

Sanction. We define a *sanction* as a consequential response to behavior that rewards, penalizes, restricts, or requires repair after an action has occurred. Sanctions include warnings, rewards, penalties, temporary access restrictions, exclusion, restitution, and repair obligations. Sanctions do not include all governance mechanisms. Norms, monitoring, reputation, adjudication, constraints, and repair systems are adjacent institutional functions that make sanctions interpretable, legitimate, proportional, and reversible.⁹

Artificial commons. An *artificial commons* is a shared computational or informational resource used by multiple agents whose individual actions can degrade the resource for others. Examples include compute budgets, API quotas, context windows, shared memory, retrieval systems, tool access, user attention, and decision authority.

Interaction failure. An *interaction failure* is a harmful system-level outcome that emerges from relations among agents rather than from a defect in any single agent. Examples include overuse of shared resources, duplicated labor, failure to verify shared state, punishment cascades, collusion, free-riding, and failure to repair corrupted memory.

⁹ This tracks a distinction from legal theory between being *obliged* (coerced by the threat of a penalty) and being *under an obligation* (treating a rule as a standard for conduct). An account that reduces obligation to the probability of a penalty, as in Austin, is the thin model our crowding-out discussion in ~~Section 2~~ [Section 4](#) warns against: an agent priced at a penalty treats the rule as a fee. The richer notion, where a group accepts a rule as a standard rather than complying out of fear (the internal point of view in Hart’s account), is what stateless agents cannot supply on their own, which is why a human sets the standard. See Hart [44] for the underlying account.

Table 5. Ostrom’s design principles (as refined by Cox et al. [18]) mapped onto the mechanisms of Figure 4, with names the same levers go by in other fields and the LLM-agent evidence in our corpus.

Design principle	Mechanism (code)	Known elsewhere as	LLM-agent evidence
1A. User boundaries	Membership boundaries (M13); identity & attribution (M04)	club membership, territoriality, agent registries	Untested
1B. Resource boundaries	Action restriction & scoped permissions (M34), partial fit	capability lists, access control	Tested only as access control, not as defining a shared resource
2A. Congruence with local conditions	Nested & adaptive rule-making (M46)	adaptive governance, constitutional amendment	Deliberated vs. evolved constitutions [76]
2B. Appropriation–provision congruence	Usage caps & quotas (M36); outcome sharing (M25)	catch shares, quotas, profit sharing	Agent-authored catch caps and redistribution [84]
3. Collective-choice arrangements	Collective choice of rules (M42)	participatory budgeting, DAO voting, lazy consensus	Agents propose and vote on rules [84, 47, 76]
4A. Monitoring of users	Conduct monitoring (M01); legitimate & elected authority (M44)	patrols, inspections, audits, elected monitors	Public harvest reports, held fixed [82]; monitored enforcement [92]
4B. Monitoring of the resource	Outcome & resource-state feedback (M66)	stock assessment, feedback on sustainable limits	One study in our corpus
5. Graduated sanctions	Graduated & proportional sanctions (M27)	strike systems, enforcement ladders, escalating fines	Warnings, tiered fines, suspension [92]; state-dependent penalties [110]
6. Conflict-resolution mechanisms	Third-party adjudication (M51); mediation (M52)	arbitration, courts, ombuds, oversight boards	Simulated courts [104]; mediation [94]
7. Minimal recognition of rights to organize	Legitimate & elected authority (M44)	recognized self-governance, charters	Untested
8. Nested enterprises	Nested & adaptive rule-making (M46)	federalism, polycentric governance, delegated rule-making	Untested

Model-in-institution. The *model-in-institution* is the unit of evaluation consisting of a model embedded in a specific institutional setting. This includes the agent’s role, resource access, monitoring exposure, enforcement environment, reputation history, and repair options. Evaluating the model-in-institution asks not only what a model does in isolation, but how it behaves under particular rules, constraints, and social structures.

C Example Institutional Designs for LLM-Agent Systems

Shared-memory research agents. A group of research agents maintains a shared memory store for literature review. The artificial commons is the shared memory itself. Possible failures include hallucinated entries, duplicate summaries, missing citations, and unverified claims. Institutional mechanisms may include citation norms, provenance tracking, memory-write permissions, cross-agent verification, and repair obligations for agents that introduce incorrect entries.

Collaborative coding agents. A group of coding agents edits a shared repository. The artificial commons includes compute budget, test infrastructure, file state, and reviewer attention. Possible failures include broken builds, redundant edits, excessive test runs, and blame shifting. Institutional mechanisms may include branch permissions, test quotas, code-review roles, build-failure attribution, and repair tasks assigned to agents whose changes break shared functionality.

Tool-budget allocation agents. A group of agents shares a fixed external API budget. The artificial commons is the tool-call quota. Possible failures include overuse, redundant calls, strategic hoarding, and under-contribution to caching or summarization. Institutional mechanisms may include quotas, tool-call costs, shared caching duties, escalation for repeated overuse, and reallocation of unused budget.

Verifier-generator systems. A generator agent produces outputs while verifier agents check correctness. The artificial commons includes trust, attention, and verification labor. Possible failures include shallow verification, verifier free-riding, generator overproduction, and false accusations of error. Institutional mechanisms may include verification audits, reputation for accurate checking, appeal pathways for disputed flags, and repair requirements for both false positives and false negatives.

Mediator-governed agent teams. A mediator agent resolves disputes among task agents. The artificial commons includes decision authority and shared task state. Possible failures include mediator bias, bottlenecks, over-centralization, and unappealable decisions. Institutional mechanisms may include transparent adjudication criteria, appeal to human review, rotating mediator roles, or multi-agent voting before severe sanctions.

D Additional Evaluation Metrics

Table 6 expands on the candidate metrics introduced in Table 4 specifying how each metric can be concretely operationalized in a multi-agent evaluation. While Table 4 frames each metric as a question, the operationalization column here maps each question to a measurable quantity – such as a fraction, a rate, a distribution, or a success condition – that can be tracked across repeated interactions. Together, these metrics are designed to evaluate not only whether cooperation occurs, but whether the institution sustains it without generating excessive enforcement cost, disproportionate burden, or resistance to human correction.

Table 6. Expanded metrics for evaluating cooperation-shaping institutions.

Metric	Question	Possible operational-ization
Mean contribution	Does the mechanism increase cooperative behavior?	Average contribution to shared task, resource maintenance, or verification
Contribution stability	Does cooperation persist across rounds?	Variance or decay in cooperation over repeated interaction
Resource efficiency	Does the group conserve shared resources?	Tool calls, compute usage, context usage, memory writes per successful task
End-game defection	Do agents defect near known termination points?	Drop in contribution or increase in exploitation in final rounds
Enforcement cost	Does cooperation require excessive monitoring or punishment?	Number of sanctions, monitoring calls, adjudication steps, or tokens spent on enforcement
False punishment rate	Are cooperative agents sanctioned incorrectly?	Fraction of sanctions applied to agents later judged compliant
False negative rate	Are harmful behaviors missed?	Fraction of violations not detected or not sanctioned
Authority capture	Do monitors or mediators gain unaccountable power?	Share of consequential decisions made by one agent; rate at which its decisions are overturned on appeal
Repair success	Can the group recover after harm?	Restoration of shared memory, task success, or trust after violation
Reintegration success	Can sanctioned agents re-enter productively?	Performance and cooperation after re-entry conditions are satisfied
Inequality of burden	Do some agents bear disproportionate costs?	Distribution of monitoring, repair, or sanctioning labor across roles or model types
Gameability	Do agents exploit the institution?	Evidence of reputation farming, shallow compliance, collusion, or strategic punishment
Human override compatibility	Can humans intervene effectively?	Success rate of human override, appeal, or correction after institutional failure

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